National
Qualifications
2023

X819/76/11 Design and Manufacture

TUESDAY, 30 MAY
1:30 PM – 3:45 PM

Total marks — 80

SECTION 1 — 25 marks

Attempt ALL questions.

SECTION 2 — 55 marks

Attempt ALL questions.

Write your answers clearly in the answer booklet provided. In the answer booklet you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

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X8197611

SECTION 1 — 25 marks

Attempt ALL questions

1. Two pedal go-karts are shown below with product information.

Commercial go-kart

Materials

plastic coated tubular mild steel frame
galvanised mild steel wheels
solid textured polypropylene seat
PVC steering wheel



Additional details

hand operated lockable brake
air filled tyres
front and rear reflectors
adjustable seat position
Price - £549.99



Domestic toy go-kart

Materials

tubular mild steel frame (painted)
hollow ABS wheels
solid polypropylene seat
PVC steering wheel
polystyrene mudguards and front panel

Additional details

hand operated brake
self-assembly required
Price - £89.99

MARKS

1. (continued)

(a) Explain why the materials chosen are suitable for these products. **6**

(*You must give **six** different explanations*.)

(b) Name **three** appropriate manufacturing processes used in the production of the go-karts **and** explain why each one is suitable. **6**

(c) Describe how the following have influenced the design of the go-karts

- function
- safety. **5**

(*To attract full marks, you must reference both function and safety in your answer.*)

(d) Describe how aesthetics has influenced the design of the go-karts. **4**

(*You must refer to a **range** of aesthetic aspects*.)

(e) Explain the benefits and drawbacks for the **manufacturer** of using standard components during the production of the go-karts. **4**

[Turn over

MARKS

SECTION 2 — 55 marks

Attempt ALL questions

2. A carabiner used in mountaineering and adventure sports is shown below.





(a) Identify a suitable material that could be used for the carabiner **and** explain why it is appropriate. **3**

The carabiner has been manufactured using the process of drop forging.

(b) Explain why drop forging is a suitable process for the production of products such as the carabiner. **2**

CAD modelling was used during the design of the carabiner.

(c) Explain the benefits of using CAD modelling during the design of the carabiner. **2**

MARKS

3. A car interior is shown below.



(a) Describe how ergonomics has influenced the design of the car interior. **6**

(*To attract full marks, you must cover all **three** aspects of ergonomics in your answer.*)

Ergonomists play a lead role in the design of products.

(b) Describe the role of **two other** members of a design team. **4**

Sub-contractors can be used in the development of products.

(c) Explain one benefit **and** one drawback of using sub-contractors. **2**

[Turn over

MARKS

4. Technology push and market pull often influence products.

(a) Describe how products with which you are familiar have been influenced by:

(i) Technology push. **2**

(ii) Market pull. **2**

Market research is carried out as part of the design process.

(b) Describe **two** appropriate methods of carrying out research into the needs of the target market. **4**

Planned obsolescence is often incorporated into products.

(c) Describe what is meant by planned obsolescence **and** give an example of how this may have influenced the design of products. **2**

5. Swimming goggles are shown below.



The seal and the strap are manufactured from elastomers.

(a) Explain why elastomers are suitable for these parts. **2**

Production and planning systems are used to improve efficiency when manufacturing products.

(b) Describe how production and planning systems are used to improve efficiency during the manufacture of products. **4**

Companies use a variety of marketing techniques to maintain or improve their market share.

(c) Describe the marketing techniques that companies could use to maintain or improve their market share. **2**

MARKS

6. Different methods can be used to identify materials in products.

(a) Describe methods that could be used to identify materials. **3**

Different assembly methods are used in the commercial manufacture of products.

(b) Outline the factors that would influence the choice of assembly method used. **3**

Manufacturers can use recycled or recyclable materials to reduce the impact of their products on the environment.

(c) Describe **other** ways in which designers and manufacturers can reduce the environmental impact of their products. **4**

7. Designers use a variety of graphic techniques throughout the design process.

Explain why a variety of graphic techniques are used at different stages of the design process. **8**

(You should reference different types of graphic techniques and where they are used in the design process.)

[END OF QUESTION PAPER]

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Question 1 Image of commercial go-kart is taken from https://mytoys.scene7.com/is/image/myToys/ext/11204940-02.jpg?$rtf_mt_prod-mainzoom_xl$

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